Topic 10: Gmail Integration – Test Cases and Results

1. Testing Overview

Workflow: Topic 10 Gmail Integration Workflow

Repository: n8n-gmail-integration

Testing Objective: Verify that n8n can connect to Gmail and send personalized plain-text emails using dynamic customer information.

2. Test Environment

- Platform: n8n Cloud
- Email Service: Gmail
- Authentication: Gmail OAuth2
- Trigger: Manual Trigger
- Data Node: Mock Customer Data
- Gmail Node: Send Personalized Email

3. Test Data

Field	Value
customerName	Alice Smith
customerEmail	User-accessible email configured privately in n8n
Email Type	Text
Subject	Welcome to N8N Course, Alice Smith!

4. Test Cases and Results

Test ID	Test Scenario	Expected Result	Actual Result	Status

TC-001	Connect Gmail OAuth2 credentials	Gmail account connects successfully	Google sign-in completed and Gmail account connected	PASS
TC-002	Configure Gmail Send operation	Resource Message and Operation Send	Gmail Send a message action configured	PASS
TC-003	Configure dynamic recipient	Recipient uses customerEmail expression	`{{ $json.customerEmail }}` configured successfully	PASS
TC-004	Configure personalized subject	Subject includes customerName	Preview displayed Welcome to N8N Course, Alice Smith!	PASS
TC-005	Configure personalized email body	Body includes personalized greeting and congratulations message	Plain-text message preview matched trainer requirements	PASS
TC-006	Execute Gmail send operation	Gmail sends email and returns message details	Gmail returned message ID, thread ID, and SENT label	PASS
TC-007	Verify email delivery	Email appears in recipient inbox	Updated email delivery verified in recipient inbox	PASS
TC-008	Execute complete workflow	All three workflow nodes execute successfully	Complete workflow executed successfully with all three nodes green	PASS

5. Latest Gmail Execution Evidence

None


```
[
  {
    "id": "1a1cfb235060892b",
    "threadId": "1a1cfb235060892b",
    "labelIds": [
      "SENT"
    ]
  }
]
```

6. Final Test Summary

Metric	Result
Total test cases	8
Passed	8
Failed	0
Overall result	PASS

7. Conclusion

All eight documented validation checks passed based on the observed configuration, execution output, and delivery verification.

The workflow successfully demonstrated Gmail OAuth2 authentication, dynamic expressions, personalized email creation, and automated email delivery through n8n.